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[GS PAPER I — GEOGRAPHY • GS PAPER II — GOVERNANCE • GS PAPER III — AGRICULTURE, ENVIRONMENT AND DISASTER MANAGEMENT]

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Deficit Monsoon, Widespread Impact



Explore how deficient monsoons impact India's agriculture, crop productivity, reservoir storage, and water security, along with sustainable strategies to strengthen agricultural resilience.

Deficit Monsoon, Widespread Impact

UPSC areas:

GS Paper I — Geography

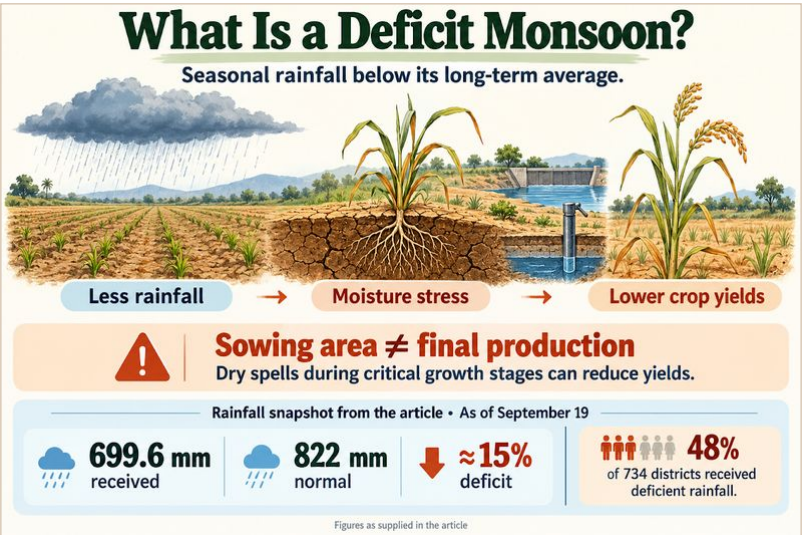
GS Paper II — Governance

GS Paper III — Agriculture, Environment and Disaster Management

WHAT IS DEFICIT MONSOON?

A deficit monsoon occurs when seasonal rainfall falls below its long-term average, reducing water availability and potentially affecting agriculture, reservoir levels, groundwater recharge, ecosystems and livelihoods across the affected region.

Deficient monsoon means increased moisture stress for crops, which will have a direct adverse impact on crop productivity. Sowing acreage alone does not reflect crop health, as prolonged dry spells during critical growth stages suppress yields. Therefore, what is projected as production now based on sowing area may not turn out to be true after harvesting operations are completed.

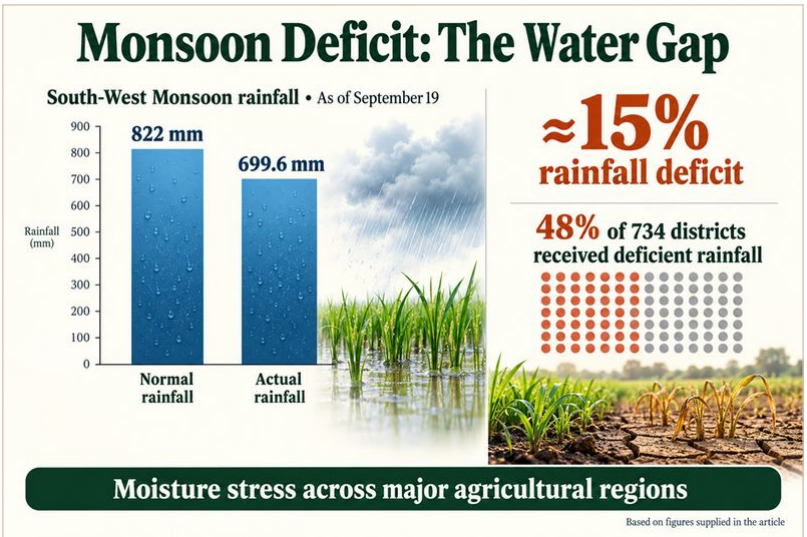


Deficient Monsoon: The Need to Assess Agricultural Impact

The India Meteorological Department's (IMD) prediction for the South-West Monsoon (SWM) turned out to be remarkably correct. Since a 15 per cent deficient monsoon compared to the Long Period Average (LPA) was severe (experienced only twice times during the last 35 years) and almost all the leading agricultural States experienced a deficit substantially higher than the national average, there is an urgent need to assess the impact of this deficient monsoon on the agricultural front.

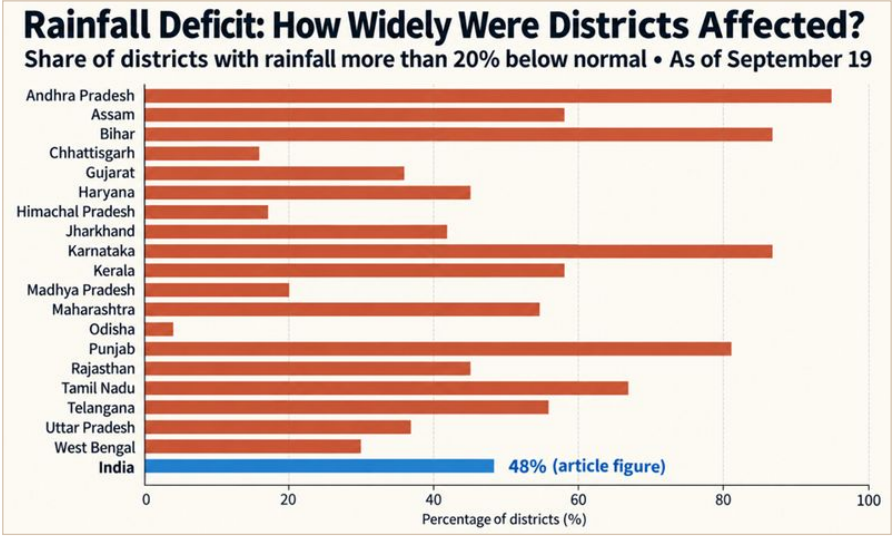
Rainfall Deficit Across Major Agricultural Regions

As per the IMD data, the country received a total rainfall of 699.6 mm from SWM as of September 19, which is much less than the normal level of 822 mm. Further, 48 per cent of the total 734 districts across the country received deficient rainfall (less than 20 of normal). This wide geographic coverage indicates that moisture stress is not restricted to a few isolated pockets, but spans major agricultural regions .



Declining Reservoir Storage

A deficient monsoon has severely affected the storage capacity of major reservoirs. As per the Central Water Commission's (CWC) weekly status of 178 major reservoirs ending October 15, only 39 per cent of them were holding more than 80 per cent of their live storage capacity. The total storage in these major reservoirs stood at 126.7 billion cubic metres (BCM) or 68.6 per cent of the total storage capacity of 184.8 BCM. This is 24 per cent less compared to last year's storage level.



The Central Water Commission is India's technical organisation overseeing water resource development, flood forecasting, reservoir monitoring and interstate water coordination.

Regional Disparities in Water Availability

This storage level is relatively better in the Western, Central and Eastern regions. Storage in the Southern region was the worst, at 44.3 per cent of total capacity. The Northern region also has less in live storage. This foretells severe irrigation shortages for the southern States in particular over the coming months.



PAST EPISODES

The impact of a deficient monsoon is expected to be severe, though the full extent will only become clear after the harvest. A similar level of deficient monsoon was experienced only in two of the last 35 years namely 2002 and 2009. The 2002 drought brought down the kharif production of foodgrains by about 22.2 per cent, rice by 21.7 per cent, nutri-cereals 25.4 per cent, pulses 14.2 per cent and oilseeds by 32.1 per cent over its previous year.

2002 Drought: Impact on Kharif Production

Decline compared with the previous year (%)

Crop / Category	Production decline
Foodgrains (overall)	22.2%
Rice	21.7%
Nutri-cereals	25.4%
Pulses	14.2%
Oilseeds	32.1%

In 2009, the decline of these crops was in the range of 10.3 per cent to 16.7 per cent over its previous year. This precedent demonstrates how sensitive India's crop output remains to truant rainfall.

Risks to Paddy and Sugarcane Production

States cultivating paddy intensively have experienced large deficits in monsoon rainfall this season. As per the Ministry of Agriculture data (as of September 4), the overall kharif sowing was 2 per cent lower than last year, but for paddy, this reduction was 3.7 per cent compared to last year. The rainfall deficit was in the range of 21-43 per cent in most intensive paddy-growing States such as Assam, Andhra, Punjab, Haryana, Telangana and Tamil Nadu. Therefore, it is highly likely that the production of paddy would drop considerably during the current harvest cycle, exceeding the 10 mt drop projected by the rice industry experts. Along with paddy, the water-intensive sugarcane crop might not meet its production target because of low water storage levels in reservoirs in States where this crop is largely cultivated.

India's Reservoir Storage: Regional Comparison

As of October 15, 2026

Region	Reservoirs	Live capacity (BCM)	Current storage (BCM)	Last-year storage (BCM)	Capacity filled (%)	Major states
Northern	11	19.8	11.4	17.3	57.5	Himachal Pradesh, Punjab, Rajasthan
Eastern	31	21.9	17.5	18.0	79.9	Bihar, Jharkhand, Odisha, West Bengal
Western	58	38.6	31.2	37.6	80.7	Gujarat, Maharashtra
Central	28	48.6	42.0	44.9	86.4	Chhattisgarh, Madhya Pradesh, Uttar Pradesh
Southern	50	55.9	24.8	50.1	44.3	Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, Telangana
All India	178	184.8	126.7	167.9	68.6	—

BCM = billion cubic metres

Source: Central Water Commission (as reproduced in supplied clipping)

Southern region has the lowest share of capacity filled.

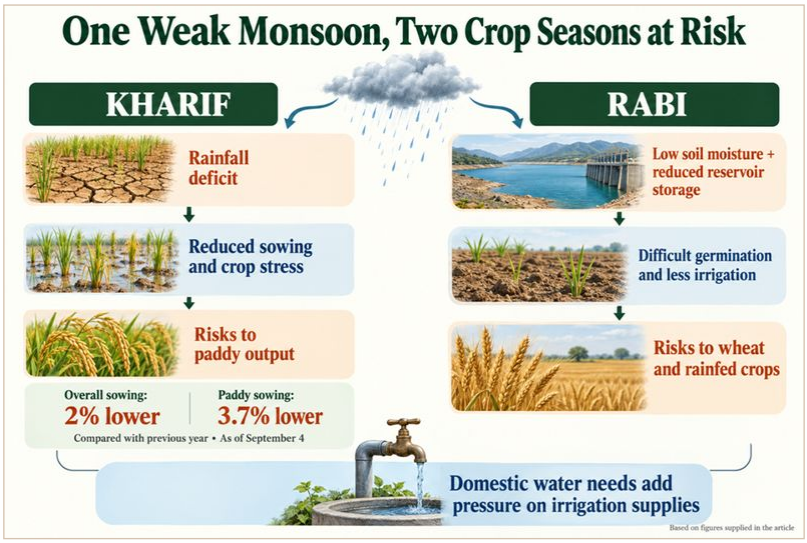
RABI PROSPECTS

A deficient SWM will also have a serious impact on the rabi crop food production target of 178 mt. Reduced soil moisture availability at the end of the SWM season will impact many rainfed rabi crops during seed germination and early growth phases. Furthermore, reduced storage of water in reservoirs will also impact the cultivation of irrigated crops such as wheat, a major crop of the rabi season. One should wait and see how rabi area cultivation will play out across different regions in the coming months.

Competing Demands for Limited Water

A deficient monsoon will not only affect agriculture but will also percolate into domestic water supply until June 2027, when the new South-West Monsoon will commence. More irrigation water from reservoirs will be diverted to meet domestic and other purposes, creating further stress for agriculture. Both the State and Central governments must

make concerted efforts to tackle the impact of this deficient monsoon.



Promoting Less Water-Intensive and Short-Duration Crops

First, given the limited water storage in major reservoirs, less water-intensive crops and short-duration crops ought to be actively promoted among farming communities with appropriate incentives. Encouraging farmers to shift away from long-duration, water-intensive varieties in water-stressed districts can prevent crop failure in the rabi season and conserve water resources.

Expanding Micro-Irrigation

Second, water-intensive crops should be cultivated using micro-irrigation systems such as drip and sprinkler technologies, which can save about 50 per cent of water with a significant increase in crop productivity.

Developing Moisture-Stress-Tolerant Crop Varieties

Third, the unpredictability of the monsoon has been increasing in recent years because of El Nino and other climatic reasons. Considering this growing rainfall instability, crop varieties that can withstand moisture stress need to be developed on a priority basis.

Strengthening Procurement and Long-Term Water Security

Fourth, the government should step up the procurement for less water-consuming crops such as oilseeds, pulses and nutri-cereals so that farmers will be attracted towards these environmentally sustainable crops. Providing an assured minimum support price and robust procurement infrastructure for low-water-consuming crops will offer farmers the necessary economic security to shift away from water-guzzling crops. Both Central and State governments must develop long-term strategies to harvest more output from less water, safeguarding the rural livelihoods and ensuring national food security in the years ahead.



Mains question: 15 marks (250 words), GS Paper III.

Examine how deficient monsoons affect India's agriculture and water security, suggesting measures for building resilience.